SwasthyaSaathi

The AI front door to Indian healthcare

WHO WE ARE

An AI-native primary-care startup building the vernacular triage layer that sits above India's fragmented care system — routing every patient to the **right specialist, at the right urgency, in their own language**, before a single rupee is misspent.

INDUSTRY	GEOGRAPHY	WE ARE
Healthcare	**India · Tier 2/3 first**	**The solver — AI-first startup**

OUR WEDGE — WHERE WE START

Fever / respiratory + diabetes / hypertension triage, in Hindi & Marathi, on WhatsApp, seeded through ASHA workers in 2-3 Tier-2/3 districts. **Highest volume, highest mis-routing, zero download friction.**

WHAT WE BUILD — IN ONE LINE

One vernacular chat → an urgency call + the right specialist + a booked verified provider. *Free to the patient, always. We earn on the referral, not on the sick.*

THE MARKET WE SIT ON TOP OF

₹6.3 L cr	India's annual health expenditure
62%	paid out-of-pocket · 85% incl. borrowing
276 M+	govt tele-triage consults already served
1.8 M	doctor/nurse shortfall vs WHO density

PROBLEM & PAIN × ₹ · USERS · WHY GENAI · SOLUTION · HOW WE BUILD · TECH · METRICS · UNIT ECONOMICS · GTM & PROFIT

The Problem — And The Three Pain Points We Attack First

WHY THE JOURNEY BREAKS · PAIN POINTS PRIORITISED BY BUSINESS IMPACT · RICE-SCORED

The system is stretched

80%
specialist shortfall at rural community health centres

Health Dynamics of India 2022-23

62%
of health spending is out-of-pocket — 85% once borrowing is counted

India Fit Report 2025

₹31,500
avg. rural hospitalisation cost; ~95% paid out-of-pocket

NSO 80th round, 2025

14%
medical inflation — highest in Asia, outpacing income growth

Industry estimates, 2025

What a patient actually does

1. **Symptom appears**
   "Is this serious? Which doctor do I even see?" — no trusted first opinion in their language.
2. **Self-medicates or asks a chemist**
   Irrational drug use → missed diagnosis, delay, resistance.
3. **Guesses a specialist / GP**
   Wrong specialist first = repeat consult + repeat diagnostics, all out-of-pocket.
4. **Delays until it's urgent**
   TB patient-journey studies show 3+ provider visits before a correct diagnosis.
5. **Reaches right care — late and poorer**
   Higher cost, worse outcome, deeper distrust of the whole system.

The three pain points we attack first

1 Wrong first specialist
Patients guess who to see → repeat consults + repeat diagnostics, all out-of-pocket.

₹ **₹500-1,500 wasted per wrong-routing episode × millions of episodes / yr**

2 No vernacular first opinion
English-only apps exclude the ~90% who would use a symptom checker in their own language.

₹ **eSanjeevani validated AI triage in 12 Indian languages for exactly this reason**

3 Dangerous delay
Self-medication + travel friction delay diagnosis until conditions turn critical.

₹ **Delay converts a ₹200 OPD problem into a ₹31,500 hospitalisation**

Why these three, in this order — RICE scored

Candidate pain point	Reach	Impact	Conf.	Effort	RICE
1. Wrong first specialist	9.0	2.5	0.6	2.0	**10.8**
2. No vernacular opinion	9.5	2.0	0.6	2.5	**8.6**
3. Dangerous delay	7.0	3.0	0.5	2.0	**7.4**

RICE = (Reach × Impact × Confidence) ÷ Effort. All scores are founder estimates (not yet market-tested) used to force a ranking, not precise measurements.

What the delay actually costs — one episode, compounding

Day 0
₹0
Chemist self-medication

→

Day 3
₹500
Wrong specialist + travel

→

Day 9
₹1,800
Second opinion + repeat tests

→

Day 20+
₹31,500
Hospitalisation · 95% OOP

THE INSIGHT THIS DECK RESTS ON — AND OUR BUSINESS GOAL
A ₹200 problem becomes ₹31,500 because nobody told the patient which door to walk through. **So we capture the top of the funnel: every correctly-routed patient = a lead + a cost saved.**

Who We Serve, and Where Today's Apps Abandon Them

USER TYPES & PERSONAS · AND WHERE TODAY'S APPS ABANDON THEM

First-time patient
Tier 2/3 & rural — low trust, low health literacy, pays cash

~65% of episodes

Family caregiver
The household's health decision-maker — usually a woman, on one shared phone

Decides for 4-6 people

Chronic / repeat patient
Diabetes, hypertension, TB — needs to know when a new symptom breaks pattern

Highest LTV segment

Time-poor self-payer
Urban working adult — has provider access, but can't afford a half-day to find out

Skips care until it's urgent



Meena, 34
Homemaker & caregiver · Nashik (Tier-2)

"When my son has fever at night, I don't know if it's normal or an emergency — so I just wait."

JTBD: "Tell me in Marathi, in 2 minutes, whether this can wait till morning."

GOALS
- Know fast if a symptom is serious
- Avoid a wasted ₹500 wrong-doctor visit
- Ask in Marathi, not English forms

PAIN POINTS
- No trusted first opinion at home
- Chemist self-medication is her default
- 2-3 hr travel to reach a specialist

Phone-first · WhatsApp-native · low English literacy

Both personas share one moment: the 60 seconds after a symptom appears, with no trusted voice to ask.



Arjun, 31
Software engineer · Bengaluru (urban)

I know something's off, but I can't burn half a workday finding out if it's nothing

JTBD: Tell me in 2 minutes if I can push through today or need to see someone now

GOALS
- Get a fast verdict without visiting in person
- Skip the queue for something minor
- Know exactly which specialist, no trial-and-error

PAIN POINTS
- Booking apps show slots, not urgency
- No way to ask "is this serious?" before committing time
- Ends up self-diagnosing via search instead

Has access · Has no time

Both personas share one moment: the 60 seconds after a symptom appears, with no trusted voice to ask.

Why today's apps miss them

Every player optimises the booking. Nobody owns the decision that comes before it. ^

	Vern.	Free	Triage	Route
Practo	✗	✗	✗	~
Tata 1mg	✗	~	✗	✗
Apollo 24|7	✗	✗	~	✗
eSanjeevani	✓	✓	✓	✗
SwasthyaSaathi	✓	✓	✓	✓

THE GAP WE WALK INTO

eSanjeevani proved vernacular AI triage works at 276 M+ consults — but it stops at advice. **Nobody routes the patient to a bookable, verified provider and closes the loop.**

^ Assessment based on our review of public apps as of 19/7/26

Why Now, and Why This Genuinely Needs GenAI

TIMING & TECHNOLOGY JUSTIFICATION · INCLUDING WHY A DECISION TREE FAILS

WHY NOW

Vernacular smartphones everywhere

Cheap data + regional-language keyboards put an AI front door in every pocket, including Tier 3 & rural.

Rails already laid

Ayushman Bharat Digital Mission health IDs, UPI payments, and eSanjeevani prove patients will use digital care.

Costs at breaking point

14% medical inflation + 85% out-of-pocket/borrowed spend means patients are actively hunting for ways not to waste money.

Supply can't catch up

The 1.8M workforce gap won't close for a decade — the only near-term lever is routing existing supply better.

WHY GENAI

A rule-based symptom checker cannot do this job. GenAI specifically unlocks:

Free-text, any-language input

Patients describe symptoms in Hinglish, Marathi, or spoken voice — GenAI parses messy natural language a decision tree can't.

Conversational follow-up

It asks the next clarifying question dynamically ('since when? any fever?') instead of forcing a fixed form.

Differential + department routing

Generates a ranked 'likely areas + which specialist' with plain-language reasoning, like eSanjeevani's validated CDSS.

Safety framing, not diagnosis

GenAI handles nuance and follow-up; a deterministic rule floor — not the model — guarantees safety

The test we held ourselves to: what breaks if we build this with rules instead of GenAI?

Real patient input	Rule-based symptom checker	GenAI triage engine	Why it decides the outcome
"do din se bukhaar aur khaansi"	✗ No match — unparsed Hinglish	✓ Parses → fever 2d + cough	Session abandoned vs completed
Voice note in Marathi	✗ Needs a typed English form	✓ ASR → intent → slots	Excludes Ramesh entirely
"chest pain" + vague context	~ Fires one generic branch	✓ Asks: radiating? sweating?	Emergency caught vs missed
Symptom overlap (dengue / typhoid)	✗ First-match wins	✓ Ranked differential + why	Right specialist, first time

The Solution: SwasthyaSaathi Triage Copilot

SOLUTION REASONING & FLOW · WIREFRAME ANNOTATED · PROTOTYPE LINKED BELOW

BEFORE

Patient guesses a doctor, self-medicates, visits 2-3 wrong providers, pays out-of-pocket each time, and is diagnosed late — at 150× the cost.

AFTER

One vernacular chat → an urgency call + the right specialist + a booked verified provider, before any money is misspent.

How it works — one conversation, four moves

1 **Describe**
Patient types or speaks symptoms in any Indian language — 'do din se bukhaar aur khaansi'. No app download, no form.

2 **Triage**
GenAI asks smart follow-ups, then returns urgency (self-care / see-doctor / emergency) + a ranked likely specialty.

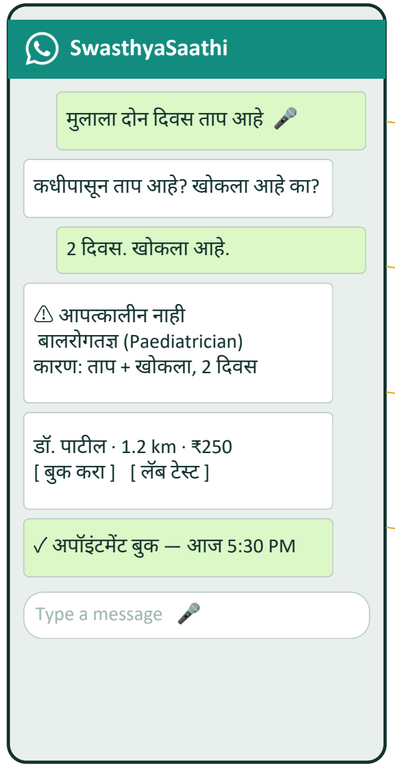
3 **Route**
Matches to the right verified specialist, nearby clinic, or teleconsult — with a plain-language 'why this doctor'.

4 **Act**
Books the consult, orders the right lab test, or escalates a red-flag case to a human doctor instantly.

Guidance, not diagnosis. A deterministic red-flag rules engine runs ***before*** the LLM and force-escalates emergencies to a human. The model never prescribes, never names a disease as fact, and always states its confidence and the next safe step. Safety is a rule, never a probability.

What Meena actually sees

Wireframe — WhatsApp, zero download. Numbered regions map to the four moves.



1 Voice or text, Marathi — no form, no English

2 Dynamic follow-up fills the missing slots

3 Urgency + specialty + a plain-language why

4 Verified provider, price, distance → booked

Booking flow shown is target-state; Part B prototype demonstrates stages 1–2 live

PART B — LIVE PROTOTYPE: the four moves above, running end-to-end on a WhatsApp sandbox number in Hindi & Marathi → Click the link and Scan the QR Code for the Prototype

How We Actually Build It

BUILD PLAN · THE TWO HARD PARTS: THE VERNACULAR FRONT DOOR + THE TRIAGE BRAIN

① The vernacular front door

How a patient talks to us at all — no app, no form, no English.

Ingress — WhatsApp Cloud API

A webhook on Meta's Cloud API takes text, voice notes, or IVR for feature phones. Zero download.

User-initiated replies are ₹0 inside the 24h service window

Speech → text (Indic ASR)

Bhashini (govt, 22 languages) primary, Whisper fine-tuned on Indic speech as fallback.

Voice is the unlock for low-literacy users

Language ID + normalisation

Detect Hinglish / Marathi / Hindi, transliterate, normalise to one canonical symptom string.

Every raw message becomes training data

Meta Cloud API · Bhashini ASR · FastAPI webhook · Redis sessions

② The triage brain

How GenAI turns messy speech into a specialty + an urgency — safely.

4 Red-flag gate — deterministic, PRE-LLM

Regex on danger phrases (chest pain+sweating, seizure, blue lips, bleeding) → hard-coded escalation, no model. Target: <0.1% miss-rate, validated pre-launch.

▸ ***Safety is a rule, never a probability***

5 Symptom extraction → structured slots

LLM returns strict JSON: {symptoms[], onset, severity, age, comorbidities}. Empty slots trigger the next question.

▸ ***This is what a decision tree cannot do***

6 RAG over the symptom→specialty KB

Vector search on a curated KB: MoHFW treatment guidelines + ICD-11 map + eSanjeevani specialty mapping.

▸ ***Grounded + citable = auditable, not hallucinated***

7 Constrained generation + schema validation

Must validate: 1 of 34 specialties + 1 of 3 urgency levels + confidence + a plain 'why'. Invalid → retry → 'see a GP'.

▸ ***The model can never invent a new answer***

LLM API → fine-tune · pgvector RAG · Pydantic schema · Rules engine

Build vs buy — what we own

Layer	Decision	Why
WhatsApp transport	BUY	Commodity · ₹0 in window
Indic ASR	BUY	Bhashini · govt · 22 langs
Base LLM	RENT	No capital at seed
Red-flag rules	BUILD	Safety can't be a model
Symptom→specialty KB	BUILD	This is the moat
Provider graph + booking	BUILD	Owns the monetisation
Payments	BUY	UPI · commodity

We rent the intelligence. We own the judgement.

What could kill the build → how we de-risk

✗ **Model mis-routes a real emergency**
✓ Deterministic gate pre-LLM + doctor audit on all red flags

✗ **Indic ASR fails on dialects**
✓ Bhashini + Whisper fallback; typed text always allowed

✗ **No verified providers in a district**
✓ Sign the lab chain first — labs outnumber specialists

PART B SCOPE: stages ①–② running end-to-end on a WhatsApp sandbox — Hindi + Marathi, fever/respiratory + diabetes/hypertension, red-flag gate live. → **[Under Construction]**

The Tech: How the Assistant Asks Good Questions

THE QUESTION ENGINE · THE ACTUAL PROMPT · DATA MODEL · CORE APIs

It's a loop, not a form

Every turn it re-decides one thing: do I know enough to route — or what single question would tell me the most?

1. Parse the turn → update the slot state
2. Re-score the top-5 specialties (RAG on the KB)
3. Run red-flag rules — every turn, first
4. Confident (> 0.75) or 4 questions asked? → answer
5. Else: ask the highest-information-gain question

Never re-asks what it knows · 1 question/turn · max 4 · in-language

The prompt behind the question

```
SYSTEM
You route patients. You never diagnose.

State:      {slots_json}
Candidates: {top5_specialty_scores}
Asked:      {n} of 4

Choose ONE action:
 ASK    — only if one question would change
          which specialty ranks first.
 ANSWER — if top score > 0.75 or n = 4.

If ASK: return the question that most
splits Candidates. Max 12 words, plain
{language}, no jargon. Never ask what
State already contains.

If ANSWER: JSON only —
{specialty, urgency, confidence, why_plain}
```

Why THIS question?

After turn 1 — state: fever · 2 days · age 6

Paediatrics .41 · Gen Med .28
Pulmonology .19 · Infectious .12

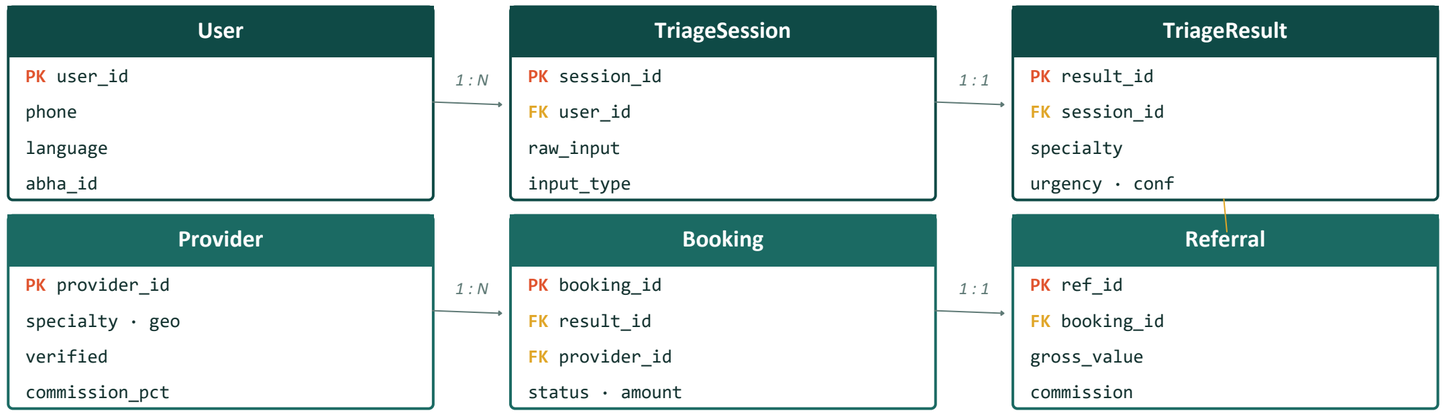
✗ **"How are you feeling?"**
Moves no score. Pure filler.

✗ **"Do you have a headache?"**
~70% say yes — barely splits the four.

✓ **"Cough or breathing trouble?"**
yes → Pulmonology .52 · no → Paediatrics .68

Either answer decides the route. That's the test — and it's why Meena answers 2 questions, not 12.

Concise ERD — capture chain feeds the fulfilment chain



PK primary FK foreign **Capture chain** = the triage. **Fulfilment chain** = the money.

Core APIs

POST	/webhook/whatsapp	inbound text / voice / IVR
POST	/triage/analyze	symptoms → urgency + specialty
POST	/triage/followup	returns the next best question
POST	/escalate/redflag	emergency → human doctor
GET	/providers/match	specialty + geo → ranked list
POST	/bookings/create	books consult; writes referral
GET	/user/history	chronic-care continuity
GET	/admin/accuracy	routed vs doctor's final call

p95 end-to-end < 4 s — fast enough that Meena doesn't put the phone down.

How We'll Know It's Working

METRICS · AAARRR FRAMEWORK · EVERY METRIC HAS A FORMULA, A TARGET AND A DIRECTION

★ NORTH STAR

Correctly-Routed Consultations per week (CRC/wk) — patients who completed a consult with the specialty we recommended, confirmed appropriate by the attending doctor. It is the only number that is simultaneously the patient's outcome, our quality score, and our revenue. ↑

S	C	R	B	A	=
Sessions started	Completion rate	Routing rate	Booking conversion	Routing accuracy	CRC / week
40,000/wk	65%	88%	38%	86%	7,494
M12 target	*M12 target*	*M12 target*	*M12 target*	*M12 target*	`CRC/wk = S × C × R × B × A`

The funnel — AAARRR, with the formula we actually

Stage	Metric	Formula	M12 target	Dir
Acquisition	New triage sessions / wk	count(session_id) where started_at ∈ week	40,000	↑
Acquisition	CAC	(ASHA incentive + field + BSP spend) ÷ new activated users	≤ ₹55	↓
Activation	Triage completion rate	sessions reaching a recommendation ÷ sessions started	≥ 65%	↑
Activation	Time-to-recommendation	p95(t_recommend − t_first_message)	≤ 4 min	↓
Engagement	Actions per active user	(bookings + escalations + follow-ups) ÷ MAU	≥ 1.4	↑
Retention	M1 repeat-use rate	users with ≥2 sessions in 30d ÷ users with ≥1	≥ 25%	↑
Retention	Chronic-cohort churn	chronic users inactive 60d ÷ chronic cohort	≤ 8% / mo	↓
Revenue	ARPS (avg revenue / session)	Σ (attach_i × ticket_i × take_rate_i)	≥ ₹32	↑
Revenue	Verified-provider fill rate	bookings accepted ÷ bookings offered	≥ 90%	↑
Referral	k-factor	invites per user × invite→activation rate	> 0.4	↑

Guardrails — must NOT worsen

Routing accuracy — ≥ 85%
`doctor-confirmed matches ÷ routed consults`
▸ *kill-switch below 75%*

Red-flag miss rate — ≤ 0.1%
`missed true emergencies ÷ true emergencies`
▸ *any single miss → incident review*

Over-referral rate — ≤ 8%
`self-care cases sent to hospital ÷ self-care cases`
▸ *we must not clog the system we're fixing*

Harm-report rate — ≤ 1.0
`harm reports ÷ 10,000 sessions`
▸ *reported to the clinical board monthly*

A tool that grows fast but routes wrong is worse than none. Guardrails outrank the North Star — if accuracy breaks, we stop.

"Active" = sent ≥1 message to the bot in the period. Every ratio is preferred over a raw count so it benchmarks across districts of different sizes.

The Profit Engine: What One Triage Session Earns

UNIT ECONOMICS · FREE TO THE PATIENT, PROFITABLE ON THE REFERRAL

Cost to serve one session

WhatsApp service window *user-initiated = free (Meta, 2026)*	₹0.00
Utility templates × 2 *@ ₹0.115 each*	₹0.23
Indic ASR (Bhashini) *voice sessions only*	₹0.40
LLM inference (~6 turns) *cached KB, short context*	₹1.80
Vector store + infra *amortised*	₹0.35
Payment gateway	₹0.42
TOTAL VARIABLE COST	**₹3.20**

Revenue from one session (ARPS)

Revenue stream	Attach	Ticket	Take	₹/session
Teleconsult / OPD	22%	₹300	20%	**₹13.20**
Diagnostic lab referral	12%	₹650	20%	**₹15.60**
E-pharmacy fulfilment	15%	₹400	10%	**₹6.00**

`ARPS = Σ ( attach_i × ticket_i × take_rate_i )` **₹34.80**

Ticket sizes benchmarked to CareEdge FY25 diagnostics data (₹759 revenue per patient, ₹187 per test) and observed Tier-2 OPD pricing. Take rates sit inside the 15–25% band organised aggregators already pay on referred volume.

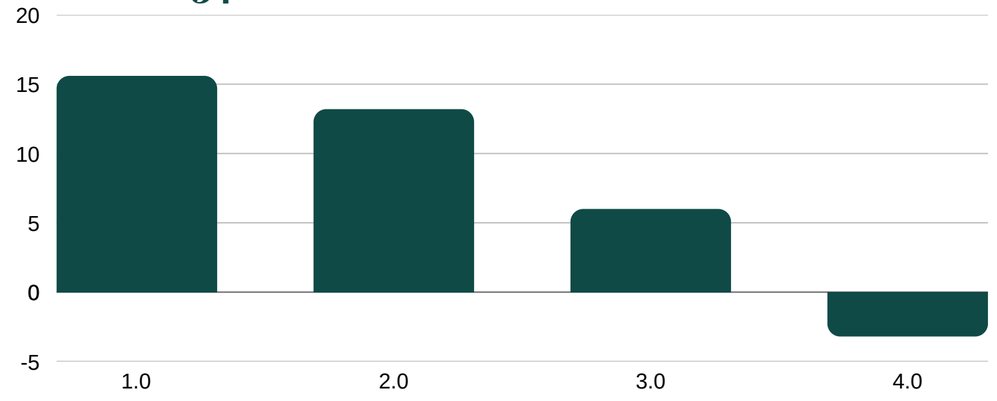
Contribution per session

ARPS	₹34.80
– Variable cost	– ₹3.20
Contribution	**₹31.60**

90.8% contribution margin

CM% = (ARPS – VC) ÷ ARPS. Software routing has almost no marginal cost — the doctor, the lab and the pharmacy carry the delivery cost, and they pay us for the demand.

Where the ₹34.80 comes from — and what it costs us



Why we don't die of CAC

The model that failed: Indian teleconsult startups paid up to **₹3,000 CAC** to earn ≤ ₹600 a consult. Structurally negative.

Our CAC	₹55	ASHA incentive ₹40 + field ₹15. No download, no ad auction.^
Our LTV	₹227	2.4 sessions/yr × 3 yrs × ₹31.60 contribution.
LTV : CAC	4.1 : 1	Payback in 1.7 sessions ≈ 8 months.

Breakeven & the margin flip

Fixed cost ₹10.0 L / month (8 FTE + infra + clinical governance)

`₹10,00,000 ÷ ₹31.60 =`

31,646 sessions / month ≈ 1,055/day ≈ 2–3 districts

PHASE 3 — THE MARGIN FLIP

An insurer pays ₹10 per triage API call to avoid one ₹31,500 mis-routed claim. Marginal cost ₹2.20 → **78% margin, zero CAC, zero delivery.**

^ LTV sensitivity (pre-launch, no cohort data yet): 1.5yr lifetime → ₹114 LTV, 2.1:1 · 2yr → ₹152, 2.8:1 · 3yr (base case) → ₹227, 4.1:1 — even the bear case clears 2:1.

Go-To-Market: Land the Funnel, Then Monetise It

GTM STRATEGY · HOW FREE TRIAGE BECOMES A PROFITABLE REFERRAL BUSINESS

PHASE 1 · LAND

Own the front door

₹0 revenue by design

Months 0–9 · 2–3 districts

- Launch free vernacular triage on WhatsApp — zero download, where users already are
- Seed via ASHA workers & local pharmacies as trust anchors (CAC ₹55, not ₹3,000)
- Sign one diagnostic-lab chain first — labs are denser than specialists in Tier-3

Goal: 1,055 sessions/day = breakeven volume

PHASE 2 · MONETISE

Turn routing into revenue

ARPS ₹34.80 · CM 90.8%

Months 9–24 · 15 districts

- Verified-provider network — 20% commission on routed consults & lab referrals
- E-pharmacy fulfilment on prescribed follow-ups (10% take)
- Household subscription — Meena's household re-engages every time a symptom appears

Goal: LTV:CAC ≥ 4:1 held while scaling 5×

PHASE 3 · SCALE

Become infrastructure

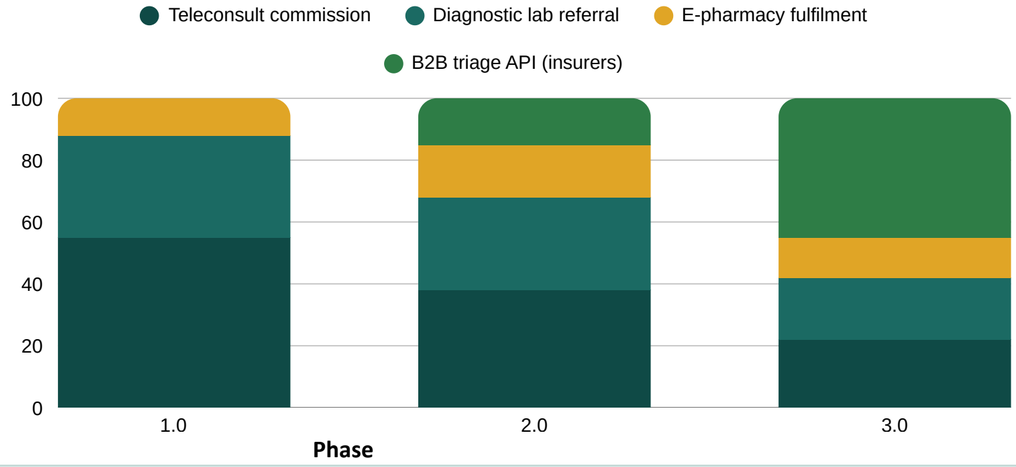
78% margin, zero CAC

Year 3+ · national

- License the triage API to insurers (cut wasteful claims) & hospital chains
- Integrate with ABDM health IDs + govt eSanjeevani as the AI layer
- Expand language & specialty coverage nationally on the same engine

Goal: 40% of revenue from B2B API

How the revenue mix shifts — and margin with it



Beachhead wedge

Start narrow — one wedge, not a platform:

Condition	Fever / respiratory + diabetes / hypertension
Language	Hindi & Marathi only
Geography	2–3 Tier-2/3 districts
Channel	WhatsApp only — no app, no web
Supply	1 lab chain + 40 verified doctors

Highest volume, highest mis-routing rate, lowest clinical risk. Widen only after routing accuracy holds ≥ 85%.

Why we defend it

Every routed episode writes one row of a proprietary India-specific **symptom → specialty → doctor-confirmed outcome** dataset. Nobody else has the outcome label — Practo knows who you booked, not whether it was right.

The flywheel: more episodes → more outcome labels → higher routing accuracy → more provider trust → more supply → more episodes.

A generic LLM can guess a specialty. It cannot know that in Nashik, this symptom cluster ends up at this doctor and gets better.

SwasthyaSaathi is the GenAI front door to Indian healthcare — vernacular triage that routes every patient to the right specialist at the right urgency, before money is wasted or diagnosis delayed. India does not need more doctors to fix this. It needs someone to tell the patient which door to walk through.